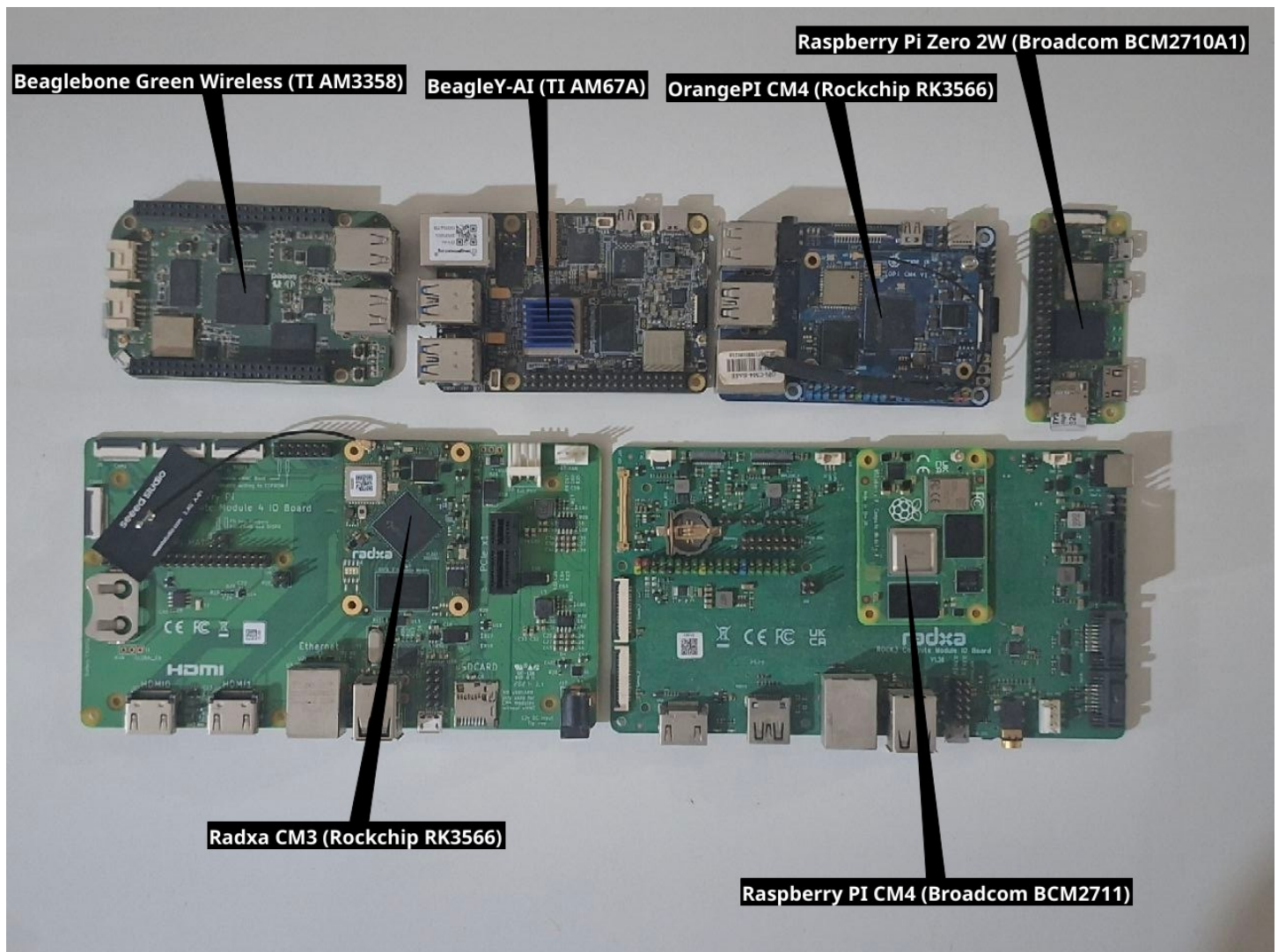


NAZIB ABRAR

Technical Supplement: Projects, Platforms, and Verification

EMBEDDED LINUX PLATFORMS

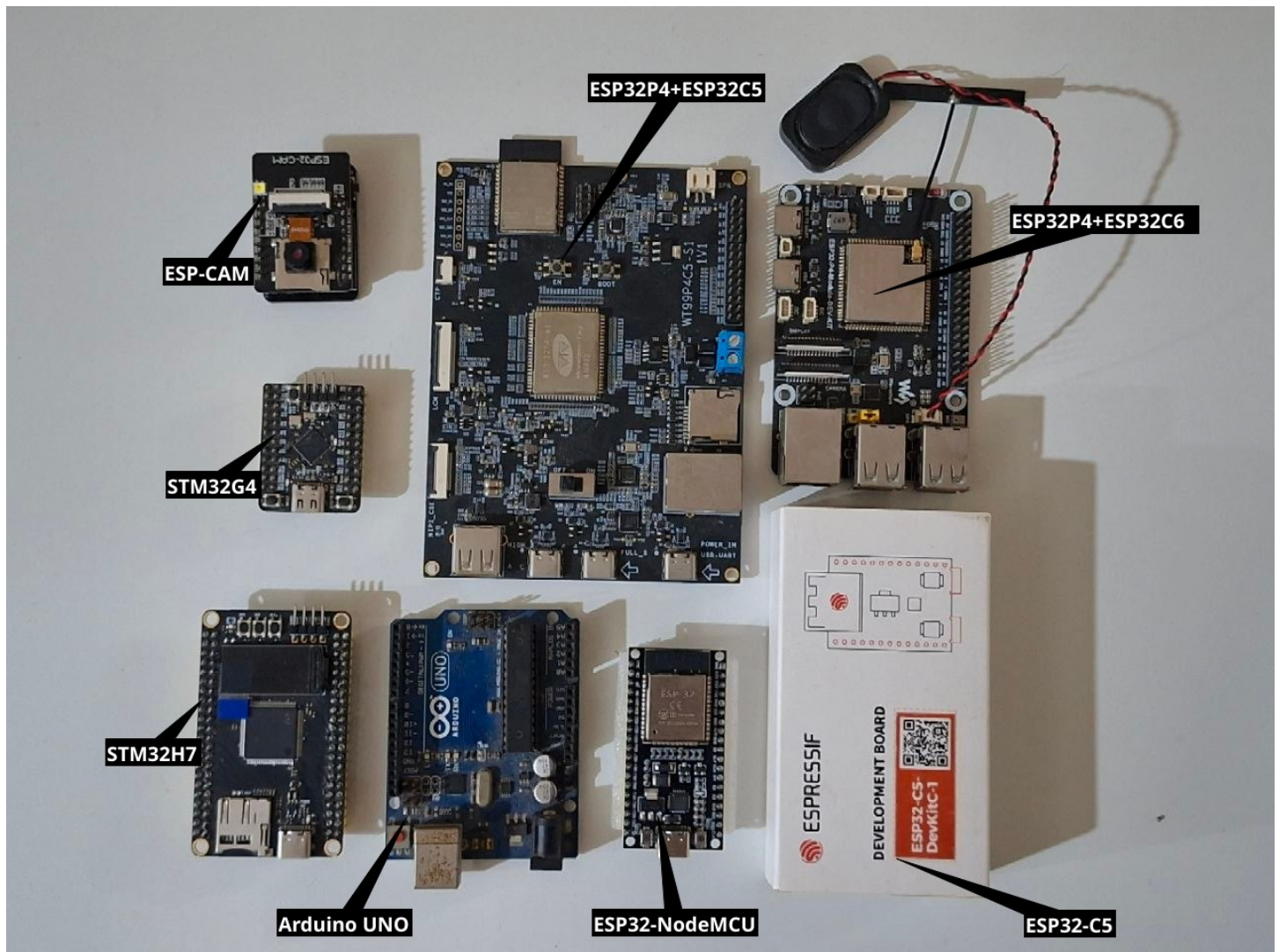
- **OrangePi CM4 and Radxa CM3** (Rockchip RK3566). Client project comprising custom carrier board development, an RKNN based computer vision stack, and a hardware accelerated GStreamer pipeline, deployed on a custom Yocto distribution built for both boards.
- **Raspberry Pi CM4 and Raspberry Pi Zero 2W** (Broadcom BCM2711 and BCM2710A1). Client projects covering a security delivery locker with WebRTC video conferencing and hardware accelerated video recording, including GStreamer pipeline development using v4l2src.
- **BeagleY-AI and BeagleBone Green Wireless** (Texas Instruments AM67A and AM3358). Conducted tutoring sessions on the Texas Instruments SDK at the RUET IoT Club, covering device driver development and the AI ISP and NPU hardware acceleration SDKs.



Single board computers and compute modules used in the projects described above.

MICROCONTROLLER PLATFORMS

- **STM32H7 and STM32G4.** Client project comprising firmware development for a CAN analyzer capable of inspecting a live CAN bus and streaming its traffic for real time data visualization.
- **ESP32-P4 with ESP32-C5, and ESP32-P4 with ESP32-C6.** Client project implementing non-AI video calling over WebRTC using esp-peer.
- **ESP32-CAM.** Employed as the camera node across a range of projects.
- **Interfaces and toolchains.** Practical experience with the SPI, I2C, and CSI camera interfaces, and firmware development in STM32CubeIDE, STM32CubeMX, ESP-IDF, and the Arduino IDE.



Microcontroller development boards used for firmware and peripheral driver work.

PROJECT DEMONSTRATIONS

Vision-Based Autonomous Drone Navigation in GPS-Denied Environments

- Research walkthrough covering the approach, perception stack, and planning: youtu.be/0HMXJcbxA3w
- Flight demonstration: youtu.be/mjQc9S_Nlv8

Robotic Arm Inverse Kinematics Control

- Inverse kinematics with visual feedback, servo driven: youtu.be/yXRN9Aov4zM
- Inverse kinematics, stepper driven: youtube.com/shorts/-wWf4OxTI5E

Line Follower Robot “Thunder” (Champion, DRMC Tech Carnival 2020)

- Short demonstration of the C++ controller software: youtube.com/shorts/ZHi7nnWHwRQ

KIBO ROBOT PROGRAMMING CHALLENGE (JAXA AND NASA)

- The Kibo-RPC is hosted by the **Japan Aerospace Exploration Agency (JAXA)** in cooperation with **NASA**. Participating teams develop software that autonomously controls **Astrobee**, NASA’s free-flying robot, inside the “Kibo” Japanese Experiment Module aboard the **International Space Station**.
- Competing as **Team Cortex Robotics** (Bangladesh), the team secured first position in the national round and **fourth position globally** in the on-orbit final, completing the mission in 3:42 with three of four correct image matches and correct identification of the target item.
- **On-orbit execution of the team’s program aboard the ISS:** youtu.be/sHPJ4LExlGM



Japan Aerospace Exploration Agency

Humans in Space

For Corporate & Research InstituteExperiment at Kibo

Rank	Country/Region	Team Name	Mission Completion Time	Number of Correct Matches	Found the correct item
1	Taiwan	iTron	3:34	4/4	OK
2	Malaysia	Automen	2:45	4/4	NG
3	Philippines	Inflection Point	4:21	4/4	NG
4	Bangladesh	Cortex Robotics	3:42	3/4	OK
5	Indonesia	Narantaka GMAT	3:11	2/4	OK
6	Japan	FUNABASHI	2:42	3/4	NG
7	USA	StarStrikers	Time Over	1/4	NG

Official JAXA final round standings.
Rank 4, Bangladesh, Cortex Robotics.

THE BUSINESS STANDARD

Saturday August 15, 2025

LatestEconomyVideosWorld+BizFeaturesSubscribe

PURSUIT

Anomno Afroz04 August, 2025, 04:05 pmLast modified: 04 August, 2025, 04:33 pm

Cortex Robotics to represent Bangladesh in global space robotics final

For a country like Bangladesh, where STEM education still faces significant setbacks, such achievement not only pushes students to dream bigger but also provides the technical framework and global exposure necessary to pursue careers in space science and robotics

TEAMCORTX ROBOTICS

Humam Hossain

Rahgir Rofi

Arif Anfin

Nazib Abrar

Ragib Rowrak

Rafsanul Haque

Collage: TBS

Cortex Robotics, a team formed by students from Bangladesh University of Textiles (BUTEX) and Rajshahi University of Engineering and Technology (RUET), has earned the honour of representing Bangladesh in the international finals of the 6th Kibo Robot Programming Challenge (Kibo-RPQ).

The challenge is a joint initiative by Jaxa (Japan Aerospace Exploration Agency) and NASA, where students programme robots to autonomously perform tasks aboard the International Space Station (ISS)

The Business Standard, 4 August 2025.
National coverage of the team’s qualification.

VERIFICATION AND CERTIFICATES

- **International Rover Design Challenge 2023.** Eleventh position globally with Team Ogrodoot (RUET), issued by the Space Robotics Society.
- **Machine Learning Specialization.** Stanford Online and DeepLearning.AI, September 2023. Verification: coursera.org/verify/specialization/T6GAYS73CU62



International Rover Design Challenge 2023, Space Robotics Society.



Machine Learning Specialization, Stanford Online and DeepLearning.AI.